



Biomedicine Content Outline

Effective September 1, 2026

Note to Candidate: This document serves as a guide to assist in examination preparation for candidates who have met NCBAHM™ eligibility requirements. Below is the content outline for the Biomedicine examination.

Please note: In regards to Clean Needle Technique (CNT), the Biomedicine module focuses on universal precautions and emergency situations in comparison to the Acupuncture with Point Location module which focuses on actual needling and its emergencies (e.g., needle angle and depth).

Domain I. Biomedical Sciences (18% of Exam)

- A. Anatomy, physiology, and pathophysiology (6%)
 - 1. Identify anatomical structures and functions of body systems from a biomedical perspective.
 - 2. Differentiate between normal and abnormal structures and functions of body systems from a biomedical perspective.
 - 3. Recognize signs, symptoms, and co-morbidities associated with common medical conditions.¹
- B. Pharmacology (7%)
 - 1. Identify common pharmaceutical categories and their functions.²
 - 2. Recognize interactions, side effects and adverse effects of pharmaceuticals.
- C. Supplements (5%)
 - 1. Identify commonly used supplements and their functions.³
 - 2. Recognize interactions, side effects and adverse effects of supplements.
 - 3. Identify clinically significant supplement – pharmaceutical interactions.

¹ See Appendix A: Medical Conditions.

² See Appendix B: Pharmaceuticals.

³ See Appendix C: Supplements.



4. Recognize signs and symptoms associated with abnormal levels of commonly used supplements.

Domain II. Patient Evaluation (26% of Exam)

A. Patient history (4%)

1. Conduct a whole-person health interview (e.g., personal health history, social history, Activities of Daily Living (ADLs), functional assessment, mental health screening).
2. Organize information obtained during interview into appropriate sections of the patient history.

B. Physical examination (14%)

1. Recognize different types of physical examinations (e.g., vital signs, orthopedic, neurological).
2. Identify appropriate application of physical examination methods (e.g., observation, auscultation, palpation).
3. Identify orthopedic examination techniques (e.g., range of motion, muscle strength testing, special tests, including orthopedic tests).
4. Identify neurological examination techniques (e.g., assessment of cognitive function, evaluation of cranial nerves, sensory and motor function, dermatomal testing, deep tendon reflexes)
5. Recognize how different types of physical examinations are performed.
6. Interpret clinically significant findings from different types of examinations.

C. Diagnostic tests (8%)

1. Identify indications for common medical imaging (e.g., x-ray, MRI, CT, PET, colonoscopy, cystoscopy, bronchoscopy, ultrasound).
2. Recognize the clinical significance of findings from medical imaging.
3. Identify indications for common laboratory tests (e.g., complete blood count, basic metabolic panel, thyroid panel).
4. Recognize the clinical significance of findings from laboratory tests.
5. Identify indications for other diagnostic tests (e.g., EMG, EKG, biopsy).



6. Recognize the clinical significance of findings from other diagnostic tests.
7. Recognize the need for preventive screening measures (e.g., cervical cancer screening).

Domain III. Analysis and Clinical Decision-making (34% of Exam)

1. Analyze assessment findings to determine their relevance to presenting complaints.¹
2. Recognize red flags (e.g., signs of stroke, heart attack, suicidal ideation, domestic abuse, trauma).
3. Establish and support the medical necessity for treatment.
4. Establish case management options (e.g., treatment, referral, co-management, discharge).
5. Differentiate between immediate, urgent, and routine referrals.
6. Recognize the potential impact of social determinants/drivers of health on treatment outcomes.
7. Recognize the potential impact of biomedical prognoses, management, and/or standard of care for common medical conditions on treatment outcomes.
8. Comprehend and utilize evidence-based information in clinical decision-making (e.g., research, expert opinion).

Domain IV. Professional Responsibilities (22% of Exam)

A. Professional and ethical conduct (5%)

1. Comply with applicable federal laws and regulations (e.g., FDA, ADA, HIPAA, CDC).
2. Comply with the legal scope of practice for Acupuncturists.
3. Adhere to the *NCBAHM™ Code of Ethics, Grounds for Professional Discipline, and the Procedures for Upholding Professional Conduct*.
4. Respect professional boundaries between patients and clinicians.
5. Recognize and respond appropriately to unsafe or unprofessional practice.
6. Treat all patients with empathy, dignity, and respect.
7. Demonstrate respect for understanding non-discriminatory practices.
8. Identify and manage potential conflict of interest situations.



9. Recognize and work within the limits of own professional competence.
 10. Terminate the course of treatment as appropriate.
- B. Infection control and workplace safety (3%)
1. Identify modes of transmission of communicable diseases (e.g., airborne, fecal-oral).
 2. Follow CDC's standard and transmission-based precautions to prevent disease transmission and minimize the risk of infection.
 3. Maintain safe clinical environment in accordance with OSHA and other standards (e.g., cleaning and disinfection of equipment, hazard communication).
 4. Comply with workplace health and safety standards.
- C. Records management and reporting (3%)
1. Document the following information in patient's chart: assessment findings, treatment plan, responses to treatment and plan for follow-up care.
 2. Maintain patient records in accordance with professional and legal standards.
 3. Maintain patient's privacy and confidentiality of protected health information in accordance with HIPAA and other legislation.
 4. Report instances of suspected abuse and neglect as required by law.
 5. Report instances of communicable disease to public health department.
- D. Insurance and billing (2%)
1. Recognize the purpose and types of professional insurance (e.g., general liability, malpractice).
 2. Differentiate between ICD and CPT medical coding systems.
 3. Describe how basic ICD and CPT (e.g., E/M) codes are used in practice.
- E. Communication and collaboration with patients, clinicians, and the public (4%)
1. Communicate clearly and be responsive to the needs of patients, clinicians, and the public.
 2. Obtain informed consent for treatment from patients as required.
 3. Work collaboratively with other clinicians in the best interests of the patient.
 4. Communicate the commonalities and differences between acupuncture and biomedicine (e.g., correlation of diagnostic categories, differences in uses of organ systems).



5. Explain acupuncture concepts to patients, clinicians, and the public using biomedical terminology.

F. Emergency management (5%)

1. Recognize situations that require special care or emergency management (e.g., seizures, falls, anaphylactic reactions, hypoglycemia).
2. Implement emergency protocols in practice (e.g., CPR).



Appendix A. Medical Conditions

The exam will focus on but may not be exclusively limited to the conditions below.

Autoimmune

- Myasthenia gravis
- Psoriasis
- Rheumatoid arthritis (RA)
- Systemic lupus erythematosus (SLE)
- Temporal arteritis

Cardiovascular

- Aneurysm
- Angina pectoris
- Arrhythmia (e.g., atrial fibrillation, premature ventricular contraction, tachycardia)
- Atherosclerosis (e.g., coronary artery disease, peripheral vascular disease)
- Blood pressure disorders (e.g., hypertension, hypotension)
- Congestive heart failure
- Deep vein thrombosis
- Dyslipidemia
- Myocardial infarction
- Postural orthostatic tachycardia syndrome (POTS)
- Raynaud's phenomenon
- Valvular disease

Dermatological

- Burns
- Contagious skin conditions (e.g., lice, fungal infections, scabies, cellulitis)
- Noncontagious skin conditions (e.g., acne, eczema, alopecia, moles)



Endocrine and Metabolic

- Adrenal disorders (e.g., Cushing's, Addison's)
- Diabetes Type 1
- Metabolic syndrome/insulin resistance (e.g., Diabetes Type 2)
- Obesity
- Parathyroid disorders
- Pituitary disorders
- Thyroid disorders (e.g., Hashimoto's thyroiditis, Graves' disease)

Gastrointestinal

- Appendicitis
- Cirrhosis
- Diverticular diseases (e.g., diverticulosis, diverticulitis)
- Food sensitivity/allergies (e.g., celiac disease, lactose intolerance)
- Gallbladder conditions (e.g., cholelithiasis, cholecystitis)
- Gastric bypass
- Gastritis
- Gastroesophageal reflux disease (GERD)
- Gastrointestinal polyps
- Hemorrhoids
- Hepatitis
- Hiatal hernia
- Inflammatory bowel disease (e.g., Crohn's disease, ulcerative colitis)
- Intestinal/biliary obstruction
- Irritable bowel syndrome (IBS)
- Pancreatitis
- Peptic ulcer (e.g., H. pylori, Campylobacter)
- Peritonitis
- Small intestinal bacterial overgrowth (SIBO)



Hematological

- Bleeding and coagulation disorders (e.g., pulmonary embolism)
- Disorders of platelets (e.g., thrombocytopenia)
- Disorders of red blood cells (e.g., anemia, polycythemia, sickle cell)
- Disorders of white blood cells (e.g., neutropenia)
- Hemochromatosis

Infectious

- Bacterial infections (e.g., E. coli, strep, MRSA, impetigo)
- Foodborne illness
- Lyme disease
- Parasitic infections
- Sexually transmitted infections
- Tuberculosis
- Viral infections (e.g., shingles, measles, influenza, post-viral illness)

Genetics

- Down syndrome
- Ehlers-Danlos syndrome (EDS)
- Marfan syndrome

Mental, Neurodevelopmental, and Behavioral

- Addictions
- Anxiety disorders
- Attention-deficit/hyperactivity disorder (ADHD)
- Eating disorders (e.g., anorexia nervosa, bulimia nervosa)
- Mood disorders (e.g., depression, bi-polar disorder)
- Obsessive-compulsive disorder (OCD)
- Post-traumatic stress disorder (PTSD)
- Schizophrenia



- Spectrum disorder (e.g., autism)
- Substance use disorder (e.g., alcohol, opioids, nicotine)
- Suicidal ideation

Musculoskeletal

- Acute, subacute, and chronic musculoskeletal conditions (e.g., trauma, degenerative disorders)
- Affecting the appendicular skeleton (e.g., meniscal injuries, frozen shoulder)
- Affecting the axial skeleton (e.g., disc herniation, TMJD)
- Bone density disorders
- Gout

Neurological

- Bell's palsy
- Concussion and traumatic brain injury (TBI)
- Dementia (e.g., Alzheimer's disease)
- Disorders of the central nervous system (e.g., cauda equina)
- Epilepsy
- Headaches (e.g., cluster, tension, migraine)
- Neuro-degenerative (e.g., multiple sclerosis, Parkinson's disease)
- Peripheral neuropathies
- Post-herpetic neuralgia
- Radiculopathies (e.g., sciatica)
- Stroke
- Transient ischemic attack (TIA)
- Trigeminal neuralgia
- Vertigo
- Other (e.g., Guillain-Barre, restless leg syndrome)



Oncological

- Cancer (e.g., organs, blood, bone, skin)
- Side effects secondary to cancer treatment (e.g., nausea, pain)

Ophthalmic and Ear, Nose and Throat

- Conditions of the ear (e.g., otitis media, tinnitus)
- Conditions of the eye (e.g., macular degeneration, conjunctivitis)
- Conditions of the nose and throat (e.g., sinusitis, allergic rhinitis)

Pulmonary

- Asthma
- Chronic obstructive pulmonary disease (COPD)
- Pneumothorax
- Respiratory tract infections (e.g., bronchitis, pneumonia)

Reproductive

- Breast conditions (e.g., lumps, mastitis)
- Erectile dysfunction (ED)
- Infertility (e.g., polycystic ovarian syndrome (PCOS), low sperm count)
- Menopause (e.g., normal and with complications)
- Menstrual and uterine disorders (e.g., dysmenorrhea, endometriosis, fibroids)
- Perinatal support (e.g., labor preparation, postpartum)
- Pregnancy (e.g., normal and with complications)
- Prostate conditions (e.g., benign prostatic hyperplasia (BPH), prostatitis)

Urinary/Renal

- Dysuria (e.g., painful urination, retention)
- Incontinence
- Infections (e.g., cystitis, pyelonephritis)
- Kidney stones



- Pelvic floor dysfunction (e.g., pelvic floor prolapse)

Miscellaneous

- Implants (e.g., pacemakers, breasts, other body part enhancements, muscles that have been treated with Botox)
- Multi-system conditions (e.g., chronic fatigue, fibromyalgia)
- Sleep disorders (e.g., narcolepsy, sleep apnea, insomnia)
- Non-dermatological fungal infections (e.g., candidiasis)



Appendix B. Pharmaceuticals

The exam will focus on but may not be exclusively limited to the list of medication categories below.

- Allergy/sinus
- Analgesic
- Anti-angina
- Anti-anxiety
- Antiasthmatic
- Antibiotic
- Anticancer
- Anticoagulant
- Antidementia
- Antidepressants
- Antidiabetic
- Antidiarrheal
- Antifungal
- Antihyperlipidemic
- Antihypertension
- Antinausea
- Anti-Parkinson
- Antipsychotics
- Antiseizure
- Antiviral
- Birth control
- Cannabinoids
- Central nervous system (CNS) stimulants/attention deficit
- Cough
- Dermatological
- Substances of abuse
- Fertility
- Gastrointestinal
- Hormonal replacement therapy
- Immune modulators/biologics
- Mood stabilizer
- Non-steroidal anti-inflammatory drugs (NSAIDs)
- Opioids
- Osteoporosis
- Sexual dysfunction
- Sleep
- Smoking cessation
- Steroids
- Stool softeners/laxatives
- Thyroid
- Weight management



Appendix C. Supplements

The exam will focus on but may not be exclusively limited to the list below.

- Amino acids (e.g., L-glutamine, L-lysine, choline)
- Anabolic supplements (e.g., creatine)
- Antioxidants (e.g., coenzyme Q10, selenium)
- Bone/Joint health (e.g., glucosamine, collagen)
- Digestive support (e.g., enzymes, probiotics)
- Energy support (e.g., ashwagandha, guarana)
- Essential fatty acids (e.g., fish oils)
- Herbal supplements (e.g., saw palmetto, valerian, turmeric)
- Homeopathic remedies (e.g., arnica, nux vomica)
- Hormonal support (e.g., melatonin, wild yam, DHEA)
- Minerals (e.g., calcium, magnesium, potassium, iron)
- Mood support (e.g., St. John's Wort, SAMe, 5-HTP)
- Sexual function support (e.g., yohimbine)
- Vitamins (e.g., A, B1-B12, C, D, E, K)



Biomedicine Examination Suggested Bibliography

The Content Outline is the primary resource for studying for this examination. The purpose of this Bibliography is only to provide the candidate with suggested resources to utilize in preparation for the examination. Candidates should feel free to consider other resources that cover the material in the Content Outline.

There is no single text recommended by NCBAHM™. All NCBAHM™ examination modules reflect practice in the United States as determined by the most recent job analysis.

NCBAHM™'s item writers and examination development committee members frequently use the following texts as resources; however, the sources used are not limited to the books listed here. The NCBAHM™ does not endorse any third-party study/preparation guide.

Anzaldúa, David. *An Acupuncturist's Guide to Medical Red Flags & Referrals*. Boulder, CO: Blue Poppy Enterprises, Inc., 2010.

Benjamin, Ben E. and Cherie Sohnen-Moe. *The Ethics of Touch: The Hands-On Practitioner's Guide to Creating a Professional, Safe, and Enduring Practice*. 3rd ed. Boulder, CO: Discovery of Books, 2020.

Bickley, Lynn S. *Bates' Guide to Physical Examination and History Taking*. 13th ed. Revised Reprint. Philadelphia: Lippincott Williams & Wilkins Publishers, 2023.

Council of Colleges of Acupuncture and Oriental Medicine. *Clean Needle Technique Manual Best Practices for Acupuncture Needle Safety and Related Procedures*. 7th ed. Council of Colleges of Acupuncture and Oriental Medicine, 2020.

[\[https://www.ccahm.org/ccaom/CNT_Manual.asp \(CCAHM Website\)\]](https://www.ccahm.org/ccaom/CNT_Manual.asp)

Jonsen, Albert R., Mark Siegler, and William Winslade. *Clinical Ethics: A Practical Approach to Ethical Decisions in Clinical Medicine*. Ninth Edition. McGraw Hill, 2022.

Kailin, David C. *Quality in Complementary and Alternative Medicine*. Corvallis, OR: CMS Press, 2006.

Magee, David J and Robert C. Manske. *Orthopedic Physical Assessment*. 7th ed. St. Louis, MO: Saunders Elsevier, 2020.

Papadakis, Maxine A., Stephen J. McPhee, and Michael W. Rabow. *Current Diagnosis and Medical Treatment*. Columbus: McGraw-Hill Education. (Current Edition)

Porter, Robert S. (Ed.). *The Merck Manual of Diagnosis and Therapy*. 20th ed. West Point, PA: Merck & Co. Inc., 2018.



Rakel, David P and Vincent Minichiello (Eds.). *Integrative Medicine*. 5th ed. Elsevier, 2023.

Stephenson, Clare. *The Complementary Therapist's Guide to Red Flags and Referrals*. 1st ed. New York: Elsevier Churchill Livingstone, 2013.

Websites

Federal regulations will take precedence over information found in other references.

Centers for Disease Control and Prevention (CDC)

<https://www.cdc.gov/>

Infectious Disease

Infection Control

First Aid

NIOSH

Centers for Medicare & Medicaid Services (CMS)

<https://www.cms.gov/>

Charting and Coding

Medline Plus

<https://medlineplus.gov/>

Drugs, Herbs and Supplements

National Center for Complementary and Integrative Health (NCCIH)

<https://nccih.nih.gov/>

Herbs at a Glance

Health Topics A-Z

[NCBAHM Code of Ethics](#)

[NCBAHM Grounds for Professional](#)

[Discipline](#) (located under Professional Discipline section)

[NCBAHM Procedure for Upholding](#)

[Professional Conduct](#) (located under Professional Discipline section)

Occupational Safety and Health Administration (OSHA)

<https://www.osha.gov/>

Universal Precautions

Office Policies and Procedures

Office of Disease Prevention and Health Promotion

<https://odphp.health.gov/>

Healthy People 2030

[Social Determinants of Health](#)

U.S. Department of Health & Human Services (HHS)

<https://www.hhs.gov/>

Medical Records

Laws and Regulations

HIPAA

National Institutes of Health, Office of Dietary Supplements

<https://ods.od.nih.gov/factsheets/list-all/>

Dietary Supplement Fact Sheets

World Health Organization (WHO)

<https://www.who.int/>

ICD Coding